

gained a lot of popularity. Players like Fitbit, Mi and Indian brands like Goqii are coming up with new fitness bands having upgraded features like real-time cardiac monitoring, blood pressure monitoring, fitness tips and so on.

This year, we came across interesting solutions from newer firms, too. US-based Nanovivo demonstrated WellSign, a fitness band that claims to be able to monitor body fat on a molecular level utilising a non-invasive photonic sensor.

At a B2B level, Tata Communications is upgrading its own wearable band, which can be used by industries to keep tabs on the health, working conditions and security of their factory workers and on-field employees.

Smart glasses are making their way in becoming a popular on-the-go data analysis platform of the future. With concepts being led by Google Glass and Microsoft HoloLens, a great number of players around the globe have come up with innovative smart glasses, many keeping in mind unique use cases.

In Japan, we came across interesting solutions like Infolinker, a pair of glasses with a fully-operational computer inside, which processes and analyses data by itself, and PicoLinker connected glasses for displaying data streamed via Bluetooth from a mobile device. These solutions were brought by Westunitis.

We also came across a miniaturised camera that can be attached to glasses, to take pictures when the user blinks in a specific manner.

A pair of smart glasses has been designed by JMACS to be worn by factory workers. It sends visuals of their real-time working conditions and challenges faced to supervisors, essentially helping the latter to become their eyes and enabling quick troubleshooting.

At Consumer Electronics Show (CES) 2018, Las Vegas, we saw smart glasses by Glatus, which can react to fatigue or drowsiness while driving, and can help hearing-impaired drivers react to sound on the road.

Among other innovative wearables, smart shoes by French firm Zhor-Tech use sensors and algorithms to detect snow and warm the feet, to detect fatigue or a disease, and to help stop driving under influence.

Automotive

Electric vehicles (EVs) and connected cars are being considered as the next-generation road riders all over the world. Design elements of automobiles are changing with larger involvement of sensors, processing units, powertrains and display screens.

Rear-view mirrors are being integrated or fully replaced by monitors, and can now be operated through controls available near the dashboard. Many of these monitors come with computer vision to help navigate or detect collision risks.

Players like Rohm Semiconductor have been working with auto manufacturers to enable these upgrades.

In Taiwan, we witnessed Askey Computer's low-power narrowband Internet of Things (NB-IoT) dashboard camera for automobiles. It provides clear road view, safety instructions to drivers on-road and real-time navigation updates to car owners.

Gadgets like Rollr Mini by Rollr, a subsidiary company of Samvardhana Motherson Group, connects to the on-board diagnostics (OBD) port of a car to provide various automotive health diagnostics and alerts in real time via a mobile application. It essentially makes a car connected!

Indian startups like HybridRickshaw have come up with an interesting automotive system to convert petrol- or diesel-based two-wheelers into hybrid or full EVs.

Bengaluru-based Okinawa Scooters have designed a battery for its two-wheelers, which can be charged in the manner similar to a smartphone battery.

Escorts Group launched India's first autonomous tractor earlier last year for farmlands. It features self-driving, precision farming through automated fertiliser and pesticide dispensing, automated safety protocols, vehicle tracking and more.

Safety Cocoon, an AI-powered automotive safe-zone concept by Sony Semiconductors, ensures vehicle safety in various driving circumstances, through 360-degree detection and risk notification.

Inova Semiconductors have introduced their upgraded ISELED automotive lighting technology at 2018

Electronica, Germany. The technology enables dynamic and intelligent back-light control. Up to 4079 digital LEDs on a flexible strip can be controlled by a single external control to generate innovative lighting applications.

Healthcare

Healthcare has been a very important area of focus with the inclusion of miniaturisation, wireless technologies, data analytics and AI. Be it for health monitoring at hospitals or for use at home, we witnessed some interesting innovations throughout the globe, by players from Taiwan, Japan, Germany and the US, among others.

Compal Electronics displayed BoostFix Posture Detector, a non-intrusive medical device that can detect posture problems, internal injuries or joint issues, and enable quick remedy.

We also witnessed how AI platforms can detect levels and conditions of diabetic retinopathy based on image analytics, developed by researchers from Industrial Technology Research Institute, Taiwan (ITRI).

We saw how a cardiac monitoring device can be used similarly to a blood pressure monitor to detect cardiac fibrillation and noises, developed by Ostar Meditech.

CardiNova, an ECG monitor as small as a bandaid, can provide ECG results on-the-fly over Bluetooth to a mobile device, which can be referred to in cardiovascular emergencies.

Another interesting solution, called Spopad, is a kind of rubber pad that clutches on the body surface and electrically stimulates the muscles to cut down excess fat automatically.

Washable smart socks have been designed for diabetic users to detect foot injuries and temperature 24x7. These keep the users alert, since high diabetes often leads to limb loss even through small injuries.

Industrial machines and test equipment

The world is powering ahead into the Industry 4.0 era, which is all about connected work floors, smart factories, automated equipment and organised utilisation of Big Data. Players like Panasonic and Fuji are promoting the idea of dark production floors.